

Building a DSR Ecosystem for Privacy-Preserving Behavioral Research: The SYNAPSIS Experience

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Privacy regulation has made the sharing and reuse of audiovisual research data a structural barrier in the social sciences and humanities. SYNAPSIS is a 24-month NWO-TDCC-SSH project applying Design Science Research methodology to address this barrier through a coordinated artifact ecosystem anchored by **MaskingOPS**—an end-to-end pipeline for privacy-preserving video de-identification and analytic utility assessment, currently under review at *Behavior Research Methods*. We derive the meta-requirements that shaped MaskingOPS from the project’s multi-stakeholder context, trace how each component of the ecosystem responds to a distinct meta-requirement, and report preliminary findings from six active or scoping pilot sites across the Netherlands and South America. Three design principles emerge from the first six months of orchestration work. We argue that the ecosystem’s contribution is not the artifacts individually but the feedback infrastructure that converts institutional constraints into transferable design knowledge.

Introduction

Behavioral researchers across the social sciences and humanities collect video as a primary data source: classroom observations, gesture corpora, clinical interaction recordings, interview video. Under the General Data Protection Regulation (European Parliament and Council 2016) and the institutional frameworks built around it, sharing or archiving this data requires prior de-identification. A researcher who wants to deposit a gesture corpus in DANS, share interview footage with a replication team, or use classroom recordings in a multi-site training study must first remove or obscure identifying facial, vocal, and body information—a task for which no shared, researcher-facing infrastructure has existed.

The gap is not primarily technical. Masking tools exist. The barrier is that those tools are designed for computer vision engineers, not for behavioral scientists who lack GPU access, are not fluent in Docker deployment, and face institutional data governance requirements that a command-line masking script cannot document. The problem is an *ecosystem* problem: the full workflow from raw video to FAIR-compliant archived output requires tool integration, training, governance templates, and community standards that no single artifact can provide.

SYNAPSIS (*Synergy Network and Platform for Inclusive Sharing of Identity-Safe AV Data*) is a 24-month NWO project funded through

the Thematic Digital Competence Center for SSH (TDCC-SSH). It applies the Design Science Research methodology (Hevner et al. 2004; Peffers et al. 2007) to build that ecosystem. This paper reports the ecosystem's artifact architecture, the meta-requirements that drove it, and preliminary findings from Month 6 of 24.

Background

DSR Ecosystems and Meta-Requirements

Design Science Research produces knowledge through the construction and evaluation of artifacts (Hevner et al. 2004). Gregor and Hevner (2013) distinguish knowledge contributions by novelty and specificity: exaptation, improvement, and invention. SYNAPSIS spans all three, since masking tools exist (exaptation), no researcher-facing end-to-end workflow exists (invention), and existing GDPR-documentation practices are improved through structured templates.

Central to our framing is the concept of *meta-requirements*: general, class-level requirements from which specific design requirements are derived (Wieringa 2014). Meta-requirements describe what any adequate artifact in a given class must achieve; they justify design decisions across components and make design knowledge transferable to analogous artifact classes. We derive four meta-requirements for the SYNAPSIS artifact class from a community survey (N = 127) and from stakeholder analysis conducted in Month 1–2.

Privacy Barriers in AV Research

An SSH-community survey conducted in Month 2 of SYNAPSIS (N = 127, distributed through CLARIAH, ODISSEI, and DARIAH Netherlands mailing lists) found that 89% of respondents held AV data they could not share due to privacy concerns, while only 23% had access to institutional masking support. Researchers who had attempted institutional review reported average DPIA durations of 4–9 months. The gap between regulatory expectation and available infrastructure is structural, not incidental.¹

¹ Survey analysis is being prepared for separate publication as part of the SYNAPSIS policy research output (M18 milestone).

The SYNAPSIS Ecosystem

Stakeholder Architecture

Four stakeholder tiers structure the ecosystem's design-evaluate-feedback cycle. *Technical partners*—Hasso Plattner Institute (Potsdam) and Netherlands eScience Center—develop and sustain the core toolchain. *Domain research partners* at Radboud University, Tilburg University, VU Amsterdam, and EUR contribute the behavioral science expertise that defines requirements and evaluates artifact fitness; co-pilot researchers at these institutions are active users

and co-evaluators, not informants. *Infrastructure partners*—SURF, DANS, and CLARIAH—provide cloud hosting, archival integration, and identity brokering. *Pilot institutions*—including Radboud Teachers Academy, University of Groningen, Tilburg University, TU Delft, Leiden, Utrecht, and the Joachim Kepler Universität Linz / Chilean courts collaboration—serve as living-lab sites, each surfacing a distinct combination of data type, IT environment, and governance constraint.

DSR Cycles

Three interlocking DSR cycles structure the project's 24-month timeline, following the process model of Peffers et al. (2007). Cycle 1 (WP1, M1–12) establishes the core artifact and proves deployment feasibility. Cycle 2 (WP2, M7–20) scales training, co-creates governance standards through the Masking Lab event, and integrates SURF cloud infrastructure. Cycle 3 (WP3, M18–24) fixes benchmarks via the Maskathon and publishes a sustainability roadmap. The co-pilot program maintains a continuous lateral channel across all three cycles, translating site-specific findings into ecosystem-level design revisions.

MaskingOPS: The Core Artifact and Its Meta-Requirements

The central artifact of the SYNAPSIS ecosystem is **MaskingOPS** (Owoyele et al. 2025)—an end-to-end pipeline for privacy-preserving video de-identification and analytic utility assessment, currently under review at *Behavior Research Methods* as a tutorial paper for behavioral scientists. MaskingOPS integrates video masking, pose-quality evaluation, and archival metadata generation into a single documented workflow. Its components directly address four meta-requirements derived from the project's stakeholder context.

Meta-Requirements

MR1 — Privacy-by-structure, not convention. Any adequate artifact must enforce privacy decisions structurally—by making it impossible to share or archive raw video without first passing through a de-identification step—rather than relying on researcher compliance with documented procedures.²

MR2 — Integrated utility assessment. Any adequate artifact must allow researchers to verify that masked data retains sufficient analytic quality for the intended research use, within the same workflow that performs the masking. Separating masking from evaluation creates an audit gap that governance frameworks cannot tolerate.

MR3 — Institutional deployability. Any adequate artifact must be deployable in the heterogeneous IT environments of Dutch research institutions—including environments without dedicated

² This requirement emerged directly from the DPIA process at Radboud Teachers Academy, where privacy officers required that the system architecture itself prevent raw-video egress, not merely that documentation attest that masking was performed.

GPU hardware, with managed-device Docker policies, and with SURFconext or CLARIAH-brokered authentication—without requiring IT customization for each site.

MR4 — Compliance legibility. Any adequate artifact must generate documentation that a privacy officer, data steward, or ethics committee can read without technical training: which masking operations were applied, to which data, under which consent regime, with what quality assessment.

Component Mapping

Each component of MaskingOPS responds to one or more meta-requirements (Table 1).

Component	Function	MR
MaskAnyone integration	Video de-identification via YOLOv11, SAM2, configurable masking strategies	MR1, MR3
MaskBench	Pose-quality evaluation across estimators and masking strategies; PCK, RMSE, kinematic stability	MR2
Workflow guides	Step-by-step researcher protocols for four deployment tiers	MR3, MR4
DPIA template	Reusable institutional privacy impact assessment framework	MR4
Consent manifest	Structured consent and data-use metadata schema	MR1, MR4
SURF cloud tier	Hosted deployment for GPU-poor institutions	MR3

Table 1: MaskingOPS components mapped to meta-requirements. MR1–MR4 are defined in §4.1.

MaskBench as Evaluation Layer

MaskBench (Riedel et al. 2025) operationalizes MR2 as a standalone evaluation framework that can be run independently or as a stage within MaskingOPS. It wraps seven pose estimators under a unified interface, computes keypoint accuracy (PCK, RMSE) and kinematic stability (velocity, acceleration, jerk) against benchmark datasets, and produces structured reports. The key finding from benchmark runs across four datasets—TragicTalkers, TED-talks, TED-kid, and ClassroomSim—is that Gaussian blurring retains the highest average keypoint accuracy (PCK = 0.86 relative to raw video) while satisfying privacy requirements, validating it as the recommended masking strategy within MaskingOPS for gesture and pose research.

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³ MaskBench is released at <https://maskbench.github.io/maskbench/> under MPL-2.0 and is indexed in the Research Software Directory.

Orchestrating Multi-Stakeholder Collaboration

The Co-Pilot Program

Each pilot institution is paired with a SYNAPSIS co-pilot who is simultaneously a platform user and an artifact reviewer. This dual role prevents the design-lag typical of conventional DSR projects, in which the gap between problem discovery and design response is measured in project cycles. Co-pilot findings are documented in a shared pilot tracker and translated into MaskingOPS revisions, training material updates, or DPIA template amendments within the same sprint.

From Constraint to Transferable Design Knowledge

The ecosystem's most important function is converting institutional constraints into documentation that other sites can use. Three constraints have produced durable design outputs.

Docker policy. Several Dutch universities block Docker on managed researcher devices. At Radboud University, this surfaced in Month 1 and forced a split-tier architecture: the researcher-facing web container runs on standard institutional VMs while the GPU compute tier runs on SURF HPC resources. This architecture is now the default MaskingOPS deployment pattern, documented in an Ansible playbook that pilot sites can run without IT customization.

DPIA latency. Institutional privacy impact assessments take 4–9 months at most Dutch universities. Rather than treating this as an external delay, SYNAPSIS developed a pre-DPIA consultation template based on the Radboud process that other sites can adapt. The University of Groningen is currently using this template for its pilot DPIA; TU Delft is adapting it following the HoMI Lab engagement in June 2026.⁴

Vocabulary mismatch. Technical documentation that is precise for an RSE is opaque for a privacy officer. MaskingOPS addresses this by producing layered documentation: a single-page researcher summary, a four-page privacy-officer brief, and a full technical specification. The Masking Lab event (October 19, 2026, VU Amsterdam) is designed around this same principle: four parallel “Masking meets X” sessions bring together masking tool developers with domain communities (FAIR data specialists, archivists, data stewards, and compute infrastructure operators) to produce joint governance outputs in vocabulary accessible to all parties.

⁴ The HoMI Lab talk at TU Delft (June 19, 2026) was the first external presentation of the full MaskingOPS + EnvisionBOX gesture challenge pipeline, reaching approximately 30 researchers in computer vision and multimodal interaction.

Preliminary Findings from Active Sites

Pilot Landscape at Month 6

Six sites are active or in scoping at Month 6 (Table 2). The diversity of institutional types, data contexts, and governance environments

is deliberate: it maximizes the range of constraints that MaskingOPS must absorb and document.

Site	Status	Key constraint surfaced
Radboud Teachers Academy	Active (M1+)	Docker policy; raw video egress control
Tilburg University (SMASHFIRE)	Active	Safety/surveillance masking; REDES-funded
University of Groningen	Technical stabilisation	VRE provisioning; CUDA version mismatch
TU Delft (HoMI Lab)	Scoping post-talk	DPIA template adaptation; gesture pipeline
Leiden University	Planning	Stakeholder identification pending
Chilean courts (JKU Linz)	Scoping call pending	Cross-jurisdictional consent; voice masking

Table 2: SYNOPSIS pilot sites at Month 6. Each site surfaces a distinctive constraint that feeds back into MaskingOPS design.

Ecosystem Enablers

Three factors have accelerated ecosystem formation beyond anticipation.

Infrastructure anchoring. SURF's participation as an infrastructure partner gave pilot institutions a credible hosting path from Day 1. Institutions unwilling to adopt a purely academic tool readily engage with a service anchored in national research infrastructure.

Publication as boundary object. The MaskingOPS tutorial paper (currently under review at BRM) circulates across stakeholder tiers in a form each can engage with: researchers read it as a methods paper; privacy officers read the compliance sections; IT staff read the deployment appendix. A single artifact serving multiple reading modes has reduced the translation work that the project team would otherwise perform one-on-one.

Open evaluation infrastructure. MaskBench's public benchmark results give privacy officers and ethics committees a quantified answer to "how much does masking degrade the data?"—a question that previously had no field-standard response. This has shortened institutional review timelines at two sites by providing reviewable evidence rather than researcher assertion.

Ecosystem Barriers

Three constraints have been harder to resolve than anticipated.

DPIA latency. Even with a reusable template, institutional review cannot be accelerated past thresholds set by governance structures outside the project's control. Pilots must train researchers before formal DPIA approval, requiring careful scope management to avoid processing real research data under a pending assessment.

GPU access fragmentation. Behavioral researchers typically lack institutional GPU allocations. SURF HPC is available to many, but onboarding adds lead time. The split-tier architecture mitigates but does not eliminate this dependency; a “Masking@Home” distributed-compute model is in early scoping as a longer-term response.

Consent-to-archive gap. Original consent forms for AV corpora rarely anticipated secondary use or de-identified archiving. Several pilot researchers hold data collected under consent regimes that do not clearly permit re-archiving even after masking. MaskingOPS’s consent manifest metadata schema is designed to make this gap explicit and auditable, but closing it requires re-consent or legal analysis that the platform cannot automate.

Training and Community Infrastructure

The training curriculum constitutes a *method* artifact in Hevner et al.’s (2004) taxonomy: a process with defined steps that produces a defined output. Two events in the immediate pipeline illustrate the curriculum’s current state.

The **Tilburg Summer School masking day** (July 16, 2026) is a full-day session within the MDIG2026 programme for researchers who have completed three days of motion tracking, pose estimation, and acoustic analysis. The session covers the MaskingOPS operations taxonomy, hands-on MaskAnyone use on participants’ own datasets, and MaskBench quality assessment. It is the first integration of MaskingOPS into an established SSH methods curriculum.

The **Masking Lab** (October 19, 2026, VU Amsterdam, 40–60 participants) is the ecosystem’s major co-creation event. The programme is set: a confirmed keynote from Roeland Ordelman (Netherlands Institute for Sound and Vision) on AV archives and CLARIAH Media Suite access; a MaskingOPS platform demonstration by Babajide Owoyele and Wessel Stoop; and four parallel “Masking meets X” sessions covering FAIR metadata (ODISSEI/DANS), digital humanities archives (CLARIAH), data stewardship and GDPR (4TU.ResearchData/TU Delft), and compute infrastructure (SURF/NLeSC). Each session runs twice so participants can attend two tracks. The designed output is a set of draft workflow protocols and FAQ documents that feed directly into MaskingOPS v2 documentation.

Design Principles

Three principles have crystallized from the first six months and are offered as provisional design knowledge in Gregor and Hevner’s (2013) sense: transferable heuristics, not laws.

DP1 — Constraints are curriculum. Institutional constraints encountered at one pilot site are valuable to all others in proportion to how systematically they are documented. A deployment failure

at one institution that is analyzed, solved, and published as an Ansible playbook step becomes a training exercise, a DPIA template section, and a deployment guide update for every subsequent site. The ecosystem's primary function is constraint accumulation and conversion, not feature delivery.

DP2 — Meta-requirements before components. Designing an artifact ecosystem without first articulating meta-requirements produces a bundle of loosely related tools. Meta-requirements serve as the architectural contract across components: they explain why each component exists, what it must achieve, and how components relate. For SYNAPSIS, articulating MR1–MR4 early made it possible to reject components that addressed only implementation-level concerns and to identify gaps (the consent-to-archive problem) that neither the platform nor the curriculum had initially addressed.

DP3 — Infrastructure anchoring enables adoption; community anchoring enables sustainability. SURF anchoring made initial institutional adoption credible. But sustainability beyond the project period depends on community ownership—open-source licensing, public benchmarks, and a developer community with intrinsic investment in the toolchain. The Maskathon (M20 target, October 2027) is designed as a transition event, moving the ecosystem from infrastructure-anchored to community-anchored.

Discussion and Open Questions

SYNAPSIS is at Month 6 of 24. Four questions will drive the next phase.

How should masking decisions be represented in archival metadata?

MaskBench defines evaluation criteria but not a vocabulary for encoding masking parameters and quality scores in deposit metadata. Developing that vocabulary is a primary agenda item for the Masking Lab and a prerequisite for the SYNAPSIS policy paper (M18 target).

What is the minimum viable DPIA scaffold? The pre-DPIA template is in use at Groningen and TU Delft. As more sites adapt it, a comparative analysis will identify which sections are genuinely portable and which must be rebuilt per institution. This is an empirical DSR question with direct policy implications.

How does MaskingOPS interact with post-hoc consent? The consent-to-archive gap affects a significant proportion of existing AV corpora. MaskingOPS can surface the gap but not resolve it. A legal and governance analysis of the options—re-consent, legitimate interest, anonymization-as-exit-from-GDPR—is needed before the policy paper can make prescriptive recommendations.

Can the Masking@Home concept reduce the GPU access barrier? A distributed-compute tier, in which GPU-poor researchers contribute small video batches to a shared processing queue, would lower the access threshold substantially. The RSSF 2026 fellowship application (submitted June 2026) targets this directly; outcome pending.

Conclusion

SYNAPSIS applies DSR to an ecosystem-level problem: the structural inadequacy of existing infrastructure for privacy-preserving AV research in the SSH. The core artifact, MaskingOPS, is designed from four explicit meta-requirements that justify every component in the pipeline and make the design knowledge transferable to analogous artifact classes. At Month 6, six pilot sites are active or in scoping, the training curriculum is integrating into established SSH methods programmes, and the Masking Lab is scheduled with a confirmed programme. The artifact is not finished; neither are the research questions. What is established is that building the ecosystem—and documenting what it resists—is itself the contribution.

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